Graphic Era deemed to be University DEHRADUN

PROJECT AND TEAM INFORMATION

Project Title

HOSPITAL MANAGEMENT SYSTEM USING MERN

Student/Team Information

Team Name: Team id:	Opticruze FS-T190
Team member 1 (Team Lead)	Lingwal,Ishank -230211112 ishanklingwal993@gmail.com

Team member 2	Tomar, Ayushi -230223592 ayushitomar624@gmail.com
Team member 3	Baliyan,Yashika-23022496 yashikabaliyan57@gmail.com
Team member 4	Tomar,Prashant-230211589 prashanttomar1645@gmail.com

PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

The project developed is a web-based application built on a technology stack known as MERN, which consists of MongoDB, Express, React, and Node. This project aims to provide a solution for managing hospital-related activities such as patient management, doctor management, appointment management, etc.

The main idea behind creating this project is to automate the hospital management system, which helps in reducing errors. This project provides a variety of features such as user management, doctor management, etc. This project helps in handling data securely

Updated Project Approach and Architecture (2 pts)

1. It was developed iteratively, adding features one by one and testing continuously.
2. It uses MERN stack architecture, which consists of MongoDB, Express.js, React.js, and Node.js.
3. It consists of three layers: front end, back end, and database.
4. React.js front end takes care of user interface and API requests.
5. Node.js/Express back end takes care of business logic and API requests.
6. MongoDB database takes care of user, doctor, and appointment data.

Tasks Completed (7 pts)

Task Completed	Team Member
1. Implemented Frontend Interface using react.js 2. Implemented Backend 3. Implemented Database Connection 4. Implemented Basic Schemas 5. Implemented User Authentication System(login/registration) 6. Integrated Frontend and Backend APIs	Yashika,Ayushi Ishank Ayushi Prashant Prashant Yashika

Challenges/Roadblocks (5 pts)

1. Difficulty was faced in integrating the front end (React) with backend APIs, particularly in handling asynchronous data.
2. It was challenging to handle MongoDB schema design along with relationships between users, doctors, and appointments.
3. It was challenging to implement authentication/authorization securely by handling tokens/sessions appropriately.
4. Faced bugs/errors during API development, which was time-consuming to debug and fix.
5. It was challenging to ensure smooth UI responsiveness.

Tasks Pending (7 pts)

Task Pending	Team Member (to complete the task)
1. Implementation of dashboard for doctor 2. Implementation of other features such as a notification system or payment system. 3. Deploying the application on a live server or cloud environment	Yashika Ayushi Ishank Prashant

Project Outcome/Deliverables

1. A fully functional Hospital Management System web application developed using the MERN stack.
2. User, Doctor, and Admin modules with role-based functionality.
3. RESTful APIs for data communication between frontend and backend.
4. A structured MongoDB database with separate collections for Users, Doctors, and Appointments.
5. User authentication system with login and registration functionality.
6. Basic UI interface for booking appointments.
7. Project documentation and report explaining the system's design and implementation.

Progress Overview

1. Completed design and planning of the "Hospital Management System."
2. Created a basic frontend interface using "React.js."
3. Created backend APIs using "Node.js" and "Express.js."
4. Set up a MongoDB database and created schemas.
5. Integrated frontend with backend for core functionality.
6. Implemented user authentication and appointment functionality.
7. Completed initial testing and debugging to get a working prototype

Codebase Information

1. The project is based on modular code, with separate folders for the frontend and backend.
2. The frontend, based on React.js, includes components, pages, and services.
3. The backend, based on Node.js and Express.js, includes routes, controllers, and middleware.
4. MongoDB schemas/models are used to define the structure of users, doctors, and appointments.
5. RESTful APIs are implemented to enable communication between the frontend and backend.
6. Git/GitHub is used to maintain version control.
7. The code is written to be scalable and maintainable.

Testing and Validation Status

Test Type	Status (Pass/Fail)	Notes
1.Unit Testing	Pass	Core modules like login and booking working
2.Integration Testing	Pass	Frontend and backend properly connected
3.API Testing (Postman)	Pass	All endpoints are returning proper responses
4.Authentication Testing	Pass	Login/Signup is working correctly
5.Data Validation Testing	Pass	Input fields are being properly validated
6.UI Testing	Pass	Input fields are being properly validated

Deliverables Progress

1Completed project design and system architecture planning.
2. Designed the frontend interface using React.js.
3. Completed backend API development using Node.js and Express.js.
4. Configured the database using MongoDB with necessary schema and collections.
5. Completed user authentication feature development (login and registration).
6. Designed core features like appointment booking and doctor management.
7. Completed initial testing and debugging of the system.
8. Deployment and advanced features development are pending for Phase 3.